

Module 23: Power BI Interview Preparation & Certification Guide

Overview

Learning Power BI is only one part of becoming a Data Analyst or Power BI Developer. The next step is proving your knowledge during interviews and demonstrating your capabilities through certifications, projects, and practical experience.

This module prepares you for Power BI interviews by covering:

- Technical interview questions
 - Scenario-based business questions
 - DAX questions
 - Data modeling questions
 - SQL questions for analysts
 - Resume preparation
 - Portfolio evaluation
 - Certification paths
 - PL-300 exam preparation
 - Career growth roadmap
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Learning Objectives

After completing this module, you will be able to:

- Answer common Power BI interview questions
 - Explain Power BI concepts clearly
 - Solve scenario-based problems
 - Prepare for PL-300 certification
 - Build a strong analytics resume
 - Present projects confidently
 - Understand career paths in BI and Analytics
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Power BI Interview Landscape

Common Roles

Data Analyst

BI Analyst

Power BI Developer

Reporting Analyst

Business Intelligence Developer

Typical Evaluation Areas

```
Power BI
  +
SQL
  +
Business Understanding
  +
Projects
```

2 Interview Process Overview

Round 1

Resume Screening

Round 2

Technical Assessment

Round 3

Project Discussion

Round 4

Managerial Interview

Round 5

HR Discussion

3 Power BI Fundamentals Questions

Q1: What is Power BI?

Answer

Power BI is Microsoft's Business Intelligence platform used to connect, transform, analyze, visualize, and share data through interactive dashboards and reports.

Q2: Components of Power BI?

Answer

- Power BI Desktop
 - Power BI Service
 - Power BI Mobile
 - Power BI Gateway
 - Power BI Report Server
-

Q3: Difference Between Dashboard and Report?

Answer

Dashboard	Report
Single Page	Multiple Pages
Summary View	Detailed Analysis
Service-Based	Desktop & Service
Executives	Analysts



Data Modeling Questions

Q1: What is a Star Schema?

Answer

A Star Schema consists of:

- Fact Table
- Dimension Tables

with the fact table at the center.

Q2: Difference Between Fact and Dimension Table?

Answer

Fact Table	Dimension Table
Stores Measures	Stores Attributes
Large Volume	Smaller Volume
Transaction Data	Descriptive Data

Q3: Why is Star Schema Preferred?

Answer

- Better Performance
 - Simpler Relationships
 - Easier DAX
 - Better Compression
-

Q4: What is a Surrogate Key?

Answer

A system-generated unique identifier used instead of business keys.

5 DAX Interview Questions

Q1: What is DAX?

Answer

DAX (Data Analysis Expressions) is the formula language used in Power BI.

Q2: Difference Between Measure and Calculated Column?

Answer

Measure	Calculated Column
Dynamic	Stored
Calculated at Query Time	Calculated at Refresh
Less Memory	More Memory

Q3: What is CALCULATE()?

Answer

CALCULATE() modifies filter context before evaluating an expression.

Q4: Explain Row Context.

Answer

Row Context means DAX evaluates one row at a time.

Q5: Explain Filter Context.

Answer

Filter Context consists of filters applied by visuals, slicers, or report filters.

Q6: Difference Between SUM and SUMX?

Answer

SUM	SUMX
Aggregation	Iterator
Faster	More Flexible
Simple Column	Row-by-Row Logic

6 Power Query Questions

Q1: What is Power Query?

Answer

Power Query is Power BI's ETL tool used for data extraction, transformation, and loading.

Q2: What Language Does Power Query Use?

Answer

M Language.

Q3: Difference Between Power Query and DAX?

Answer

Power Query	DAX
Data Preparation	Data Analysis
Before Loading	After Loading
M Language	DAX Language

7 Power BI Service Questions

Q1: What is Power BI Service?

Answer

Cloud platform for publishing, sharing, refreshing, and managing Power BI content.

Q2: What is a Workspace?

Answer

A collaborative environment containing reports, dashboards, semantic models, and apps.

Q3: What is a Gateway?

Answer

A secure bridge connecting on-premise data sources with Power BI Service.

Performance Optimization Questions

Q1: How Do You Improve Performance?

Answer

- Use Star Schema
 - Remove Unused Columns
 - Use Measures Instead of Columns
 - Optimize DAX
 - Reduce Visual Count
-

Q2: What is Performance Analyzer?

Answer

A tool that identifies report performance bottlenecks.

Q3: What is Incremental Refresh?

Answer

Refreshing only new or changed data instead of the entire dataset.

Scenario-Based Questions

Scenario 1

A dashboard loads slowly.

Approach

1. Use Performance Analyzer
 2. Review DAX
 3. Check Relationships
 4. Reduce Visuals
 5. Optimize Model
-

Scenario 2

Revenue numbers differ across reports.

Approach

1. Verify Business Definitions
 2. Review Measures
 3. Use Shared Semantic Models
 4. Standardize KPIs
-

Scenario 3

Users should only see their region.

Solution

Implement Dynamic RLS using:

```
USERPRINCIPALNAME()
```

10 SQL Questions for Power BI Interviews

Q1: Difference Between WHERE and HAVING?

Answer

WHERE	HAVING
Filters Rows	Filters Groups
Before Aggregation	After Aggregation

Q2: What is a JOIN?

Answer

Combines data from multiple tables.

Types

- INNER JOIN
- LEFT JOIN
- RIGHT JOIN
- FULL JOIN

Q3: What is a Primary Key?

Answer

Uniquely identifies a row.

Q4: What is a Foreign Key?

Answer

References a primary key in another table.

Project-Based Questions

Interviewers frequently ask:

Explain Your Project.

Structure:

Business Problem

Dataset

Data Cleaning

Modeling

DAX

Dashboard

Insights

Recommendations

Behavioral Questions

Q1

Tell me about yourself.

Q2

Describe a challenging project.

Q3

How do you handle tight deadlines?

Q4

How do you learn new technologies?

Evaluation Criteria

- Communication
 - Problem Solving
 - Teamwork
 - Adaptability
-

Resume Building

Recommended Sections

Summary

Skills

Projects

Certifications

Education

Highlight

- Power BI

- SQL
- Excel
- Python
- Data Analysis

Portfolio Review Checklist

Include

3–5 Quality Projects

GitHub Repositories

Screenshots

Documentation

Business Insights

Avoid

Incomplete Projects

Poor README Files

Generic Dashboards

PL-300 Certification

Official Certification

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Exam Focus Areas

Prepare Data

Model Data

Visualize Data

Analyze Data

Deploy & Maintain Assets

Certification Roadmap

Beginner

PL-300

Intermediate

Azure Fundamentals

Advanced

Fabric Certifications

Enterprise

Data Engineering Certifications

Career Paths

Data Analyst

Focus:

- Reporting
 - Dashboards
 - Business Insights
-

Power BI Developer

Focus:

- DAX
- Modeling
- Performance

BI Developer

Focus:

- Data Warehousing
 - ETL
 - Reporting
-

Analytics Consultant

Focus:

- Strategy
 - Business Solutions
-

Best Practices

Understand Concepts

Not just tools.

Build Real Projects

Practice SQL Daily

Learn Business Metrics

Explain Decisions

Not just features.

Common Interview Mistakes

Memorizing Answers

Weak Project Explanations

Poor DAX Knowledge

Ignoring Business Context

Lack of Portfolio Documentation



Key Terminologies

Term	Definition
DAX	Data Analysis Expressions
RLS	Row-Level Security
Semantic Model	Centralized business dataset
Star Schema	Fact-dimension model
KPI	Key Performance Indicator
Gateway	On-premise connector
Incremental Refresh	Refresh changed data only
Workspace	Collaboration area
Deployment Pipeline	Environment promotion tool
PL-300	Power BI Data Analyst certification



Mock Interview Questions

Explain CALCULATE().

Explain Row Context vs Filter Context.

Difference Between SUM and SUMX.

What is Star Schema?

What is Incremental Refresh?

How Would You Optimize a Slow Dashboard?

Explain Dynamic RLS.

Difference Between Dashboard and Report.

Explain a Power BI Project.

What Happens When You Publish a Report?

Career Capstone

Portfolio Requirements

Power BI Projects

Minimum:

5 Projects

Recommended:

8–10 Projects

Required Categories

- Sales Analytics
- HR Analytics
- Finance Analytics
- Customer Analytics
- Supply Chain Analytics

Documentation

Each project should include:

- README
- Screenshots
- Business Problem
- KPIs
- Insights
- Recommendations

GitHub

Maintain a professional portfolio repository.

Interview Readiness

Be able to explain:

- Data Model

- DAX Logic
- Business Value
- Dashboard Design Decisions



Module Summary

In this module, you learned:

- ✓ Power BI Interview Process
- ✓ Data Modeling Questions
- ✓ DAX Questions
- ✓ Power Query Questions
- ✓ Power BI Service Questions
- ✓ Performance Optimization Questions
- ✓ Scenario-Based Questions
- ✓ SQL Interview Questions
- ✓ Project Presentation
- ✓ Resume Building
- ✓ Portfolio Review
- ✓ PL-300 Certification
- ✓ Career Paths

You now have the knowledge required to prepare for Power BI interviews, build a professional portfolio, pursue certification, and transition into Data Analyst, BI Analyst, or Power BI Developer roles.



Next Module

Module 24: Microsoft Fabric & The Future of Power BI

Topics Covered:

- Introduction to Microsoft Fabric
- Fabric Architecture
- OneLake
- Data Engineering
- Data Warehousing
- Real-Time Analytics
- Lakehouses

- Fabric Integration with Power BI
- Enterprise Analytics Platforms
- Future Trends in Business Intelligence